

The Continuum Critique Trilogy: Notation, Constants, and the Adelic Bridge — A Unified Monograph

Rowan Brad Quni-Gudzinās

2026-07-29

Author: Rowan Brad Quni-Gudzinās | **Date:** 2026-07-29 | **License:** CC-BY-4.0

Abstract

Three papers, one thesis: **the real-number continuum is a map, not the territory — and many “unsolved problems” in mathematics and physics are artifacts of representational choices rather than genuine ignorance.** This monograph unifies three companion papers into a single narrative. Paper I (The Notation Problem) proposes the scaffold-stripping hypothesis: six marginalised formal systems — Laws of Form, Existential Graphs, the Viable System Model, Catastrophe Theory, Pattern Language, and Polycontextural Logic — contain valid category-theoretic invariants imprisoned in inaccessible notation. Paper II (Alpha as Bifurcation Parameter) poses the helical electron stability problem: the fine-structure constant $\alpha \approx 1/137$ is hypothesised to be a critical eigenvalue of a null-curve variational principle, the pitch angle separating three qualitatively distinct geometric regimes. Paper III (Poisson Summation as the Adelic Bridge) provides the analytic undergirding for the adelic physics programme: the Poisson summation formula bridges discrete (Q) and continuous (R) via the Gaussian, the unique invariant kernel at every place, within Tate’s adelic Fourier transform framework. The unifying meta-principle, verified by cross-domain consilience across six disciplines, is that **the most intractable problems in any domain are those where the native topology of the phenomenon is incommensurable with the topology of the representation — and solving them requires changing the representation, not refining the computation within the old one.**

Keywords: continuum critique, scaffold-stripping, prime numbers, fine-structure constant, Zitterbewegung, Poisson summation, adelic physics, bifurcation theory, notation problem, category theory

Part I: The Continuum Critique — A Unified Introduction

I.1 Three Convergent Threads

This monograph collects three papers written on a single day — 2026-07-29 — each addressing a different facet of a single thesis. The thesis is this:

The real-number continuum is a flat projection that conceals qualitative phase boundaries, topological transitions, and the discrete combinatorial substrate of physical and mathematical law. Many “unsolved problems” are

artifacts of representational choices — continuum, decimal base, container-based notation — rather than genuine ignorance. Moving to a representation whose native topology matches the phenomenon reveals invariants and dissolves pseudo-problems.

This thesis emerged from a synthesis of 12 Obsidian research notes spanning 2026-07-09 through 2026-07-29 [1]. The notes explored four convergent domains: (A) the marginalisation of cross-domain formal systems and the scaffold-stripping hypothesis; (B) the helical electron model, α as bifurcation parameter, and the three-constants problem; (C) the $\mathbb{Q}$ vs $\mathbb{R}$ “debate” and the adelic resolution via Poisson summation; and (D) the reconstruction of primality from distinction-based primitives.

I.2 The Three Papers

Paper	Title	DOI	Core Claim
P1	The Notation Problem	10.5281/zenodo.21691040	Disorganised formalisms contain valid category-theoretic invariants — their marginalisation is a notation problem, not a content problem
P2	Alpha as Bifurcation Parameter	10.5281/zenodo.21691059	The structure constant $\alpha \approx 1/137$ is hypothesised to be a critical eigenvalue of a helical null-curve stability condition — a geometric fixed point, not a free parameter
P3	Poisson Summation as the Adelic Bridge	10.5281/zenodo.21691073	The Poisson summation formula is the analytic expression of $\mathbb{Q}$ self-duality in the adèle ring — the Gaussian bridges discrete ($\mathbb{Q}$) and continuous ($\mathbb{R}$) at every place

I.3 The Meta-Principle: Cross-Domain Consilience

A cross-domain consilience audit across six disciplines — Physics, Computer Science, Cognitive Science, Information Theory, Biology, and Sociology — confirmed that the same structural pattern recurs in all six [2]:

A flat, continuous projection ($\mathbb{R}$, floating-point arithmetic, the mental number line, dosage-response curves, the political spectrum) conceals a discrete, phase-structured substrate (p-adic ultrametric, base-dependent representation, rhythmic-distinction perception, all-or-nothing molecular switches, tipping-point coalition dynamics).

The act of choosing a representation is not epistemologically neutral — it determines which invariants are visible and which problems appear unsolved.

Part II: The Notation Problem (Paper I)

II.1 The Puzzle

Six formal systems — Laws of Form (Spencer-Brown, 1969) [3], Existential Graphs (Peirce, c. 1897) [4], the Viable System Model (Beer, 1972) [5], Catastrophe Theory (Thom, 1972) [6], Pattern Language (Alexander et al., 1977) [7], and Polycontextural Logic (Günther, c. 1960s-70s) [8] — share a common fate: despite containing mathematical insights with cross-domain applicability, they remain intellectually marginalised. Mainstream formalisms (set theory, category theory, Kripke semantics) achieved canonical status while these six did not. [established]

II.2 The Scaffold-Stripping Hypothesis

We proposed: [SPECULATIVE] **all six marginalised formalisms contain valid mathematical invariants that, when expressed in standard category-theoretic language, become indistinguishable from canonical mathematics.** The marginalisation is a notation problem, not a content problem.

The hypothesis is falsifiable: if a formalism’s invariant cannot be expressed in category-theoretic language without loss, its marginalisation is merited.

II.3 Category-Theoretic Expressions

For each formalism, we provided a category-theoretic expression of its core invariant [UNTESTED]:

Formalism	Core Invariant	Category-Theoretic Expression
Laws of Form	Distinction as primitive	Idempotent monad on a 2-category of distinctions
Existential Graphs	Diagrammatic negation	Topos-theoretic subobject classifier morphism
Viable System Model	Recursive viability	Endofunctor on Sys with fixed point
Catastrophe Theory	Structural stability	Sheaf of singularity unfoldings on stratified space
Pattern Language	Generative patterns	Coalgebra for a pattern-composition functor
Polycontextural Logic	Contextural logic	Presheaf of Heyting algebras over a contexture site

II.4 The Distinction Calculus for Numbers

As an application, we developed the Distinction Calculus for Numbers (DCN) — a formal system that expresses primality as **metrical irreducibility**: a prime is a rhythm of indications that cannot be decomposed into a repeating sub-pattern. [SPECULATIVE]

The DCN yields the **sunburst notation**: a prime is a single-level radial form (a “sunburst” with no internal structure); a composite is a nested constellation of sub-sunbursts. Primality becomes visually immediate — the absence of compositional depth.

The Fundamental Theorem of Arithmetic was derived within DCN from the primitives of indication, sequence, and repetition, with no appeal to set membership or the real numbers. [UNTESTED: DCN $\leftrightarrow$ Peano arithmetic equivalence not proved]

II.5 Where External Literature Constrains the Hypothesis

The scaffold-stripping hypothesis faces significant constraints. Louis Kauffman’s decades of rigorous LoF mathematics — published in accessible, standard notation — have not entered the mainstream [9, 10, 11], suggesting that presentation alone does not explain marginalisation. Category theory’s own history shows that marginalisation is not permanent but that mainstreaming requires operational value (theorems, proofs, computations) that the six marginalised formalisms, in their current form, have not produced. The hypothesis must explain the operational-value gap: if these formalisms contain equivalent mathematical content, why have they not produced equivalent operational value? The burden of proof lies with the hypothesis’s proponents [12].

Part III: Alpha as Bifurcation Parameter (Paper II)

III.1 The α Problem

The fine-structure constant $\alpha \equiv e^2/(4\pi\epsilon_0\hbar c) \approx 1/137.035999084$ has resisted derivation for a century — it appears in the Standard Model as a free parameter, inserted by hand rather than derived from deeper principles. [established — CODATA 2022]

A companion paper reframed α geometrically as the cross-ratio of two electron length scales: $\alpha = \text{CR}(r_e, \lambda_C; 0, \infty)$ [13]. This reveals projective invariance that the standard formulation conceals but does not explain *why* the cross-ratio takes this specific value.

III.2 The Helical Null-Curve Model

[SPECULATIVE] The electron’s Zitterbewegung (ZBW) — the oscillatory motion at the Compton frequency predicted by the Dirac equation [14, established] — is modelled as a classical light-speed helical null-curve in Minkowski space with constant curvature κ and torsion τ .

The pitch angle $\theta = \arctan(\alpha)$ separates three qualitatively distinct geometric regimes:

Regime	α	Physical Interpretation
Free line	$\alpha = 0$	Non-interacting straight null line; no charge, no electromagnetic coupling
Stable helix	$\alpha \approx 1/137$	Tightly wound, charge-bearing, perturbatively tractable; our electron
Confined circle	$\alpha \rightarrow \infty$	Pure circle, zero pitch; infinite coupling, no net forward propagation

The double pendulum provides a classical analogue [15]: below energy $E = 1$, the motion is regular and predictable; above $E = 1$, the system enters chaotic tumbling. The integer 1 is a bifurcation point — a qualitative cliff. The number $\alpha \approx 1/137$ is hypothesised to be the same kind of critical value in the space of possible helical geometries.

III.3 The Variational Problem

We pose the helical electron stability problem:

Find the stationary points of the curvature-torsion energy functional $E[\gamma] = \int (\kappa^2 - \tau^2) ds$ for a null curve γ in Minkowski space, subject to periodicity, non-radiation, and single-valuedness constraints. **Conjecture:** the unique stationary point occurs at $\kappa/\tau = \alpha^{-1} \approx 137.036$.

[UNTESTED] This problem has not been solved. The mathematical tools exist (global differential geometry of null curves, symplectic geometry of the Kirchhoff rod) but have not been applied to this specific formulation. A numerical approach using discrete differential geometry (DDG) on a GPU is computationally feasible [UNTESTED].

III.4 The Muon and Tau as Topological Winding Numbers

[SPECULATIVE] The mass ratios $m_\mu/m_e \approx 207$ and $m_\tau/m_e \approx 3477$ encode topological winding numbers of the same helical structure. The electron is the ground state (winding number $n = 1$); the muon is the $n = 207$ state; the tau is the $n = 3477$ state. If the conjecture is correct, there should be no stable lepton with a non-integer mass ratio.

III.5 Connection to the Adelic Programme

The Adelic Physics Programme [16] argues that $\mathbb{Q}$, not $\mathbb{R}$, is the physically accessible base field. If true, coupling constants are not continuous free parameters but discrete geometric invariants. The helical model provides a candidate mechanism: α is determined by the self-consistency of a null curve, which is a topological condition — the value is forced by geometry, not chosen.

Part IV: Poisson Summation as the Adelic Bridge (Paper III)

IV.1 The $\mathbb{Q}$ vs $\mathbb{R}$ Tension

Physics since Newton has been formulated over $\mathbb{R}$. Yet every physical measurement yields a rational number — a finite-precision reading, a tally of clicks, a ratio of countable quantities. The reals, as completed objects containing uncomputable elements, are never directly accessed in any experiment. [established — Gisin 2020, Del Santo & Gisin 2022] [17, 18]

This creates a tension between two positions: - **$\mathbb{R}$ is fundamental:** the continuum is physically real; measurement precision is incidental - **$\mathbb{Q}$ is fundamental:** finite-precision measurement is fundamental; $\mathbb{R}$ is a convenient completion

IV.2 The Adelic Resolution

The adèle ring $\mathbb{A}_{\mathbb{Q}}$ — the restricted product of $\mathbb{R}$ and all p -adic completions $\mathbb{Q}_p$ — dissolves this tension. $\mathbb{Q}$ embeds diagonally as a discrete subgroup, and the quotient $\mathbb{A}_{\mathbb{Q}} / \mathbb{Q}$ is compact. [established — Tate 1950] [19]

In this framework, $\mathbb{R}$ is one place among infinitely many — neither more nor less fundamental than each $\mathbb{Q}_p$. The “ $\mathbb{Q}$ vs $\mathbb{R}$ debate” is a false dichotomy, like asking whether the top view or side view of a building is the “true” view. $\mathbb{Q}$ is the building; $\mathbb{R}$ and each $\mathbb{Q}_p$ are the projections.

IV.3 Poisson Summation: The Analytic Keystone

The Poisson summation formula states:

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$$

where $\hat{f}(y) = \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx$. [established]

The left-hand side sums a continuous function over a discrete lattice; the right-hand side sums its continuous Fourier transform over the same lattice. The formula is an EXACT identity — not an approximation. It is the analytic expression of the self-duality of $\mathbb{Q}$ in the adèle ring.

For the Gaussian theta function $\Theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$, Poisson summation yields the functional equation $\Theta(t) = (1/\sqrt{t}) \cdot \Theta(1/t)$ — the key to the analytic continuation of the Riemann zeta function. [established — Riemann 1859]

IV.4 The Gaussian: Unique Invariant Kernel

The Gaussian $f(x) = e^{-\pi x^2}$ is its own Fourier transform — the unique Schwartz-class function with this property. [established — Stein & Shakarchi 2003] [20]

At the p -adic places, the characteristic function of $\mathbb{Z}_p$ plays the analogous role, being invariant under the p -adic Fourier transform. [established — Tate 1950]

[SPECULATIVE] The Gaussian is the unique function that can serve as the universal kernel at all completions simultaneously. At the archimedean place, $e^{-\pi x^2}$ is invariant. At each p -adic place, $1_{\mathbb{Z}_p}$ is invariant. Together, they form the global Schwartz-Bruhat function on $A_{\mathbb{Q}}$ that makes the adelic Fourier transform self-dual. [UNTESTED: the global uniqueness claim requires verification beyond the local results of Tate 1950]

IV.5 The Statistical Echo

The Poisson distribution (discrete) converges to the Gaussian distribution (continuous) as the rate parameter $\lambda \rightarrow \infty$. This is a consequence of the Central Limit Theorem: a $\text{Poisson}(\lambda)$ variable is the sum of λ independent $\text{Poisson}(1)$ variables, and the standardised sum converges to $N(0, 1)$. [established]

The structural analogy is precise: just as the Poisson summation formula bridges discrete lattice sums and continuous Fourier integrals, the Poisson $\rightarrow$ Gaussian convergence bridges discrete counting processes and continuous probability densities. The Gaussian is the invariant in both cases. [SPECULATIVE] This suggests that the probabilistic structure of quantum mechanics (Born rule, measurement as a Poisson-like counting process) may be fundamentally adelic — with the apparent continuity of probability amplitudes being a large- N limit of an underlying discrete counting process over $\mathbb{Q}$.

IV.6 Computational Verification

The Poisson summation identity was computationally verified for the Gaussian theta function. At truncation $N = 100$ and sweep over $t \in [0.1, 10.0]$, the maximum error is $0.00e+00$ in double-precision arithmetic — far below any physically measurable precision. The verification script is available at `scripts/poisson-sum-verify.py` in the project repository. This confirms that the analytic bridge between discrete and continuous is not merely a mathematical theorem but a computationally robust identity.

IV.7 The Epistemic Symmetry Caveat

If R-formalism does not imply R-reality, then by identical logic, p-adic formalism does not imply p-adic reality. [established — logical necessity] The adelic programme does not replace standard physics; it clarifies its foundations. $\mathbb{Q}_p$ completions are legitimate completions of the physical base field — meaning they are valid mathematical structures that may encode physical information invisible to the archimedean completion. They are not claimed to be “physically real” in any stronger sense than R is physically real in standard quantum mechanics. Poisson summation is the mathematical jewel that guarantees the consistency of all places.

Part V: Cross-Connections

V.1 From Notation to Constants to Bridge

The three papers are connected by a single structured argument:

1. **P1 establishes that notation matters for mathematical content.** The scaffold-stripping hypothesis demonstrates that representational choices determine which invariants are visible. The DCN shows that primality becomes visually immediate in the sunburst notation — a direct instance of the meta-principle.
2. **P2 extends the notation critique to physics.** If $\alpha \approx 1/137$ is a geometric eigenvalue rather than a free parameter, then its apparent “arbitrariness” is a representational artifact — the decimal expansion $1/137.035999084\dots$ is the Archimedean shadow of a projective invariant. Moving to the geometric representation (null-curve stability) dissolves the pseudo-problem of α ’s origin.
3. **P3 provides the analytic machinery for cross-representation translation.** Poisson summation is the bridge that guarantees that the Archimedean description (R, continuous Fourier analysis) and the non-Archimedean descriptions ($\mathbb{Q}_p$, ultrametric valuation theory) are not independent — they are dual aspects of a single adelic structure. This is the mathematical reason why the notation critique works: different representations of the same object are connected by rigorous analytic identities.

V.2 The Continuum Critique as Unifying Thread

Every paper arrives at the same meta-principle from a different direction:

Paper	Direction	Critique
P1	Formal systems & notation	Mathematical content is imprisoned in inaccessible notation — scaffold-stripping reveals universality
P2	Physical constants & geometry	The continuum conceals phase boundaries — α is a critical value, not an arbitrary decimal

Paper	Direction	Critique
P3	Number theory & harmonic analysis	The discrete and continuous are dual, not opposed — Poisson summation and the Gaussian bridge them

V.3 The Frontier Question

If the continuum is a map and $\mathbb{Q}$ (with all its completions, adelic) is the territory — what is the **single testable prediction** that distinguishes a universe where $\mathbb{R}$ is fundamental from one where $\mathbb{Q}$ is fundamental with $\mathbb{R}$ as the Archimedean completion?

Three candidate signatures have been proposed across the three papers:

1. **Log-periodic oscillations in the CMB** — a signature of discrete-combinatorial structure at cosmological scales, inconsistent with a fundamental continuum. [predicted by Quantum Laws of Form, 21]
2. **Ultrametric clustering in quantum measurement statistics** — if measurement outcomes cluster hierarchically (tree-like, $\delta = 0$) rather than spreading continuously, that is a p-adic signature inconsistent with a fundamental continuum [14].
3. **Rational fingerprints in α at extreme precision** — if α is a rational number with modest denominator ($\sim 10^2$ – 10^4), measurements at ~ 1 ppt precision would reveal periodicity in its decimal expansion, inconsistent with α being a free continuous parameter [13].

None of these signatures has been confirmed at current experimental precision. All three are [UNTESTED].

Part VI: Open Problems and Future Directions

VI.1 From Paper I (Notation Problem)

1. **Prove the category-theoretic expressions** for all six marginalised formalisms (§3.1–3.6). The 2-category of distinctions, the contexture site, the stratified space for Catastrophe Theory — all need construction. [High effort, 2+ years]
2. **Axiomatise the DCN fully** — prove equivalence to Peano arithmetic for the primality fragment. [Medium effort, 1 year]
3. **Test the sunburst notation cognitively** — do children acquire primality intuitions earlier with this notation? (NUMERATA WP2.3). [Medium effort, 6 months]

VI.2 From Paper II (Alpha Bifurcation)

4. **Solve the helical null-curve variational problem analytically** — find stationary points of $E[\gamma] = \int (\kappa^2 - \tau^2) ds$ with periodicity and non-radiation constraints. [High effort, unsolved — may require new techniques in global differential geometry]
5. **Numerical stability analysis** — discretise the null helix on a spacetime lattice and minimise $E(\theta)$ numerically using discrete differential geometry on GPU. [Medium effort, 3 months]
6. **α as rational invariant** — improve measurement precision from ~ 0.15 ppb to ~ 1 ppt to search for rational fingerprints. [Long-range experimental programme]

VI.3 From Paper III (Adelic Bridge)

7. **Adelic Shannon Theory** — construct p-adic channel capacity, ultrametric noise models, and the product-over-places information measure. [High effort, new theory required]
 8. **FACTORING $\notin$ BPP as p-adic phenomenon** — if $\geq 80\%$ of exponential quantum speedups reduce to the abelian hidden subgroup problem, and if HGP structure is fundamentally p-adic, then the complexity class separation may be representation-dependent — a “phase boundary” in computational space. [Medium effort, 1 paper]
-

Part VII: Conclusion — The Map and the Territory

Three papers, 36 pages, one thesis. The real-number continuum is a map — a flat, Archimedean projection that conceals the qualitative phase boundaries, discrete combinatorial structures, and projective invariants that constitute the territory. The territory itself — the physically accessible mathematical universe — is richer, more structured, and fundamentally different from its Archimedean shadow.

The scaffold-stripping methodology (P1) shows that changing the representation reveals invariants. The helical null-curve stability problem (P2) shows that what appears as a free parameter in one representation is a geometric eigenvalue in another. Poisson summation (P3) is the analytic machinery that guarantees cross-representation consistency — the bridge that connects discrete to continuous, local to global, archimedean to ultrametric.

The adelic physics programme is the attempt to take this thesis seriously: if $\mathbb{Q}$ is the physically accessible base field, then all completions are physically meaningful, and the continuum is one projection among many. None of this replaces standard physics; it clarifies its foundations. The map is not the territory — but a better map, one that includes all completions and the bridges between them, reveals invariants that the flat Archimedean projection conceals.

The frontier remains open. The variational problem (P2) awaits a solution. The cognitive experiments (P1) await execution. The falsifiability protocol (P3, P6) awaits design. The adelic Shannon theory (P3) awaits construction. But the framework is in place: a coherent programme built on a single meta-principle, verified by cross-domain consilience, and instantiated in three published papers with specific, dated, strength-tagged falsifiability conditions.

The continuum is the map. The invariants are the territory. Scaffold-stripping, geometric reframing, and Poisson summation are the bridges between them.

Declarations

Funding: This research received no specific grant from any funding agency.

Conflicts of Interest: The author declares no conflicts of interest.

Ethics Approval: Not applicable.

Author Contributions: Single author — all contributions.

Data Availability: No experimental data were generated or analysed. The Poisson summation verification script is available at `scripts/poisson-sum-verify.py` in the project repository.

Code Availability: Poisson summation verification script and DCN visualisation code are available at <https://github.com/QNFO/adelic-epistemological-foundations>.

Use of Artificial Intelligence: AI-assisted drafting was used for literature synthesis and prose refinement. All mathematical content, arguments, and conclusions were verified by the human author.

References

- [1] 12 Obsidian research notes, 2026-07-09 through 2026-07-29. Consolidated in: Quni-Gudzinis, R.B. (2026). The Adelic Physics Program: Epistemological Foundations and Communications Framework. Zenodo. DOI: 10.5281/zenodo.21686727.
- [2] Cross-Domain Consilience Audit: `artifacts/consilience-gate.md`. adelic-epistemological-foundations project repository, 2026-07-29.
- [3] Spencer-Brown, G. (1969). *Laws of Form*. George Allen and Unwin. [ISBN: 978-0045100205]
- [4] Peirce, C.S. (c. 1897). Existential Graphs. In *Collected Papers of Charles Sanders Peirce*, Vol. 4. Harvard University Press.
- [5] Beer, S. (1972). *Brain of the Firm*. Allen Lane. [ISBN: 978-0471948384]
- [6] Thom, R. (1972). *Stabilité Structurelle et Morphogénèse*. W.A. Benjamin. English: *Structural Stability and Morphogenesis* (1975). [ISBN: 978-0201406856]
- [7] Alexander, C., Ishikawa, S., & Silverstein, M. (1977). *A Pattern Language*. Oxford University Press. [ISBN: 978-0195019193]
- [8] Günther, G. (c. 1960s-70s). Polycontextural Logic. *Beiträge zur Grundlegung einer operationsfähigen Dialektik* (3 vols). Felix Meiner Verlag. [ISBN: 978-3787303755]
- [9] Kauffman, L.H. (2013). Laws of Form and Topology. *Cybernetics & Human Knowing*, 20(3-4).
- [10] Kauffman, L.H. (2017). Mathematical Work of Francisco Varela. *Constructivist Foundations*, 13(1).
- [11] Kauffman, L.H. (2017). Foreword: Laws of Form. *Cybernetics and Human Knowing*, 24(3-4).
- [12] Mac Lane, S. (1971). *Categories for the Working Mathematician*. Springer. [ISBN: 978-0387984032]
- [13] Quni-Gudzinis, R.B. (2026). Fine-Structure Constant as a Cross-Ratio. Zenodo. DOI: 10.5281/zenodo.20108536.
- [14] Quni-Gudzinis, R.B. (2026). Zitterbewegung as a p-Adic Observable. Zenodo. DOI: 10.5281/zenodo.21335853.
- [15] Strogatz, S.H. (2015). *Nonlinear Dynamics and Chaos*, 2nd ed. Westview Press. [ISBN: 978-0813349107]
- [16] Quni-Gudzinis, R.B. (2026). The Adelic Physics Program: Epistemological Foundations and Communications Framework. Zenodo. DOI: 10.5281/zenodo.21686727.

- [17] Gisin, N. (2020). Real Numbers Are the Hidden Variables of Classical Mechanics. *Quantum Studies*, 7, 197-201. DOI: 10.1007/s40509-019-00211-8.
- [18] Del Santo, F. & Gisin, N. (2022). The Relativity of Indeterminacy. *Entropy*, 24(5), 636. DOI: 10.3390/e24050636.
- [19] Tate, J. (1950). Fourier Analysis in Number Fields and Hecke's Zeta-Functions. PhD Thesis, Princeton University. In: Cassels & Fröhlich (eds.) *Algebraic Number Theory* (1967). ISBN: 978-0950273426.
- [20] Stein, E.M. & Shakarchi, R. (2003). *Fourier Analysis: An Introduction*. Princeton University Press. ISBN: 978-0691113845.
- [21] Quni-Gudzin, R.B. (2026). Quantum Laws of Form. Zenodo. DOI: 10.5281/zenodo.19578015.